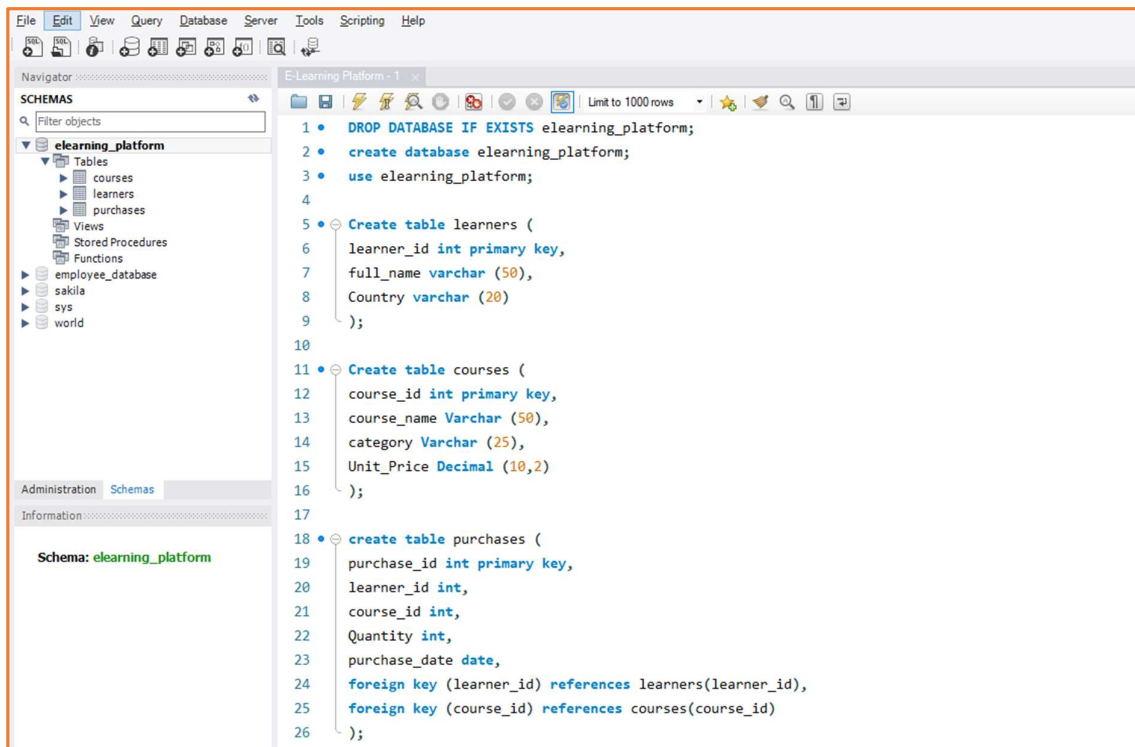


“Analysing E-Learning Platform Purchases using MySQL”

Tasks:

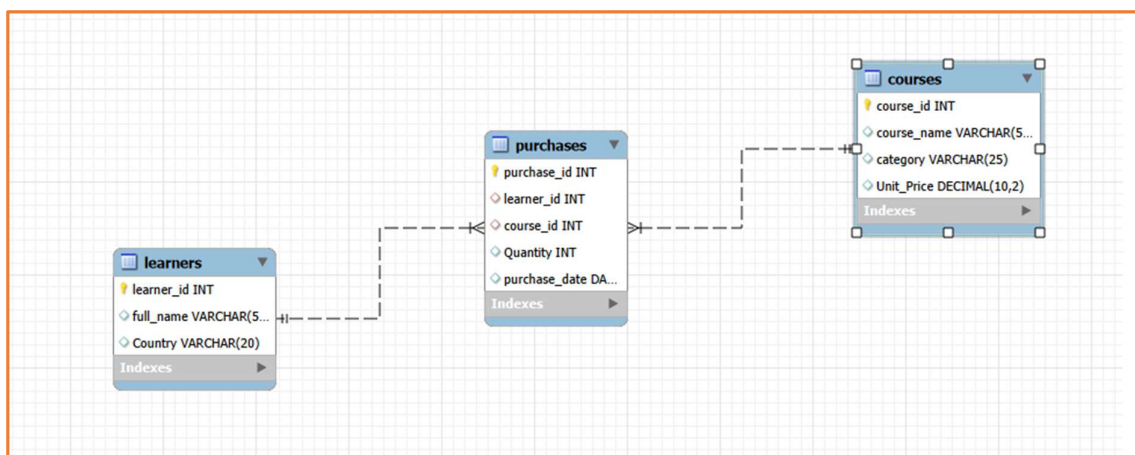
1. Create the database and schema. Populate the Schema:

- Create a Database for this project and
- Create all three tables in MySQL with appropriate data types and relationships.



The screenshot shows the MySQL Workbench interface. The left sidebar displays the 'SCHEMAS' tree with 'elearning_platform' selected. The main editor window shows the following SQL script:

```
1 • DROP DATABASE IF EXISTS elearning_platform;
2 • create database elearning_platform;
3 • use elearning_platform;
4
5 • Create table learners (
6   learner_id int primary key,
7   full_name varchar (50),
8   Country varchar (20)
9 );
10
11 • Create table courses (
12   course_id int primary key,
13   course_name Varchar (50),
14   category Varchar (25),
15   Unit_Price Decimal (10,2)
16 );
17
18 • create table purchases (
19   purchase_id int primary key,
20   learner_id int,
21   course_id int,
22   Quantity int,
23   purchase_date date,
24   foreign key (learner_id) references learners(learner_id),
25   foreign key (course_id) references courses(course_id)
26 );
```



- Insert sample data covering at least:
 - 4–5 learners
 - 4–5 courses (spread across multiple categories)
 - 6–8 purchase records

The screenshot shows a database management tool interface with a menu bar (File, Edit, View, Query, Database, Server, Tools, Scripting, Help) and a toolbar. The left sidebar contains a 'SCHEMAS' panel with a search filter and a tree view showing the database structure: elearning_platform (Tables: courses, learners, purchases; Views: Stored Procedures, Functions; employee_database, sakila, sys, world). Below this is an 'Administration' tab and an 'Information' section for the 'purchases' table, showing columns: purchase_id (int PK), learner_id (int), course_id (int), Quantity (int), and purchase_date (date).

The main editor displays the following SQL script:

```

28 • insert into learners (learner_id, full_name, Country)
29 values
30 (101, 'Aarav Sharma', 'India'),
31 (102, 'Priya Nair', 'India'),
32 (103, 'John Smith', 'USA'),
33 (104, 'Emily Davis', 'UK'),
34 (105, 'Rahul Verma', 'India'),
35 (106, 'Sophia Brown', 'Canada'),
36 (107, 'Karthik Raj', 'India'),
37 (108, 'Olivia Wilson', 'Australia');
38
39 • insert into courses (course_id, course_name, category, Unit_Price)
40 values
41 (201, 'MySQL for Beginners', 'Database', 1200.00),
42 (202, 'Advanced Excel', 'Analytics', 900.00),
43 (203, 'Python for Data Analysis', 'Programming', 1500.00),
44 (204, 'Power BI Dashboarding', 'Visualization', 1800.00),
45 (205, 'Digital Marketing Basics', 'Marketing', 1000.00),
46 (206, 'Statistics Essentials', 'Analytics', 1300.00);
47
48 • insert into purchases (purchase_id, learner_id, course_id, Quantity, purchase_date)
49 values
50 (301, 101, 201, 1, '2025-01-05'),
51 (302, 102, 203, 1, '2025-01-07'),
52 (303, 103, 202, 2, '2025-01-10'),
53 (304, 104, 204, 1, '2025-01-12'),
54 (305, 105, 205, 3, '2025-01-15'),
55 (306, 106, 206, 1, '2025-01-18'),
56 (307, 107, 201, 2, '2025-01-20'),
57 (308, 108, 203, 1, '2025-01-22'),
58 (309, 101, 204, 1, '2025-01-25'),
59 (310, 102, 202, 2, '2025-01-28'),
60 (311, 103, 205, 1, '2025-02-01'),
61 (312, 105, 206, 2, '2025-02-04');
62
63 • SELECT * FROM learners;
64 • SELECT * FROM courses;
65 • SELECT * FROM purchases;
  
```

File Edit View Query Database Server Tools Scripting Help

Navigator

Schemas

Filter objects

cleaning_platform

Tables

courses

learners

purchases

Views

Stored Procedures

Functions

employee_database

satish

sys

world

Administration Schemas

Information

Table: purchases

Columns:

purchase_id int PK

learner_id int

course_id int

Quantity int

purchase_date date

Result Grid

Filter Rows

Limit to 1000 rows

58 (309, 101, 204, 1, '2025-01-25'),

59 (310, 102, 202, 2, '2025-01-28'),

60 (311, 103, 205, 1, '2025-02-01'),

61 (312, 105, 206, 2, '2025-02-04'),

62

63 SELECT * FROM learners;

64 SELECT * FROM courses;

65 SELECT * FROM purchases;

learners

learner_id	full_name	Country
101	Aarav Sharma	India
102	Priya Nair	India
103	John Smith	USA
104	Emily Davis	UK
105	Rahul Verma	India
106	Sophia Brown	Canada
107	Karthik Raj	India
108	Olivia Wilson	Australia

courses

course_id	course_name	category	Unit_Price
201	MySQL for Beginners	Database	1200.00
202	Advanced Excel	Analytics	900.00
203	Python for Data Analysis	Programming	1500.00
204	Power BI Dashboarding	Visualization	1800.00
205	Digital Marketing Basics	Marketing	1000.00
206	Statistics Essentials	Analytics	1300.00

purchases

purchase_id	learner_id	course_id	Quantity	purchase_date
301	101	201	1	2025-01-05
302	102	203	1	2025-01-07
303	103	202	2	2025-01-10
304	104	204	1	2025-01-12
305	105	205	3	2025-01-15
306	106	206	1	2025-01-18
307	107	201	2	2025-01-20
308	108	203	1	2025-01-22
309	101	204	1	2025-01-25
310	102	202	2	2025-01-28
311	103	205	1	2025-02-01
312	105	206	2	2025-02-04

learners 16 x courses 17 purchases 18

learners 16 courses 17 purchases 18

learners 16 courses 17 purchases 18 x

Apply Revert

Output

Action Output

#	Time	Action	Message	Duration / Fetch
1	16:17:30	SELECT * FROM learners LIMIT 0, 1000	8 row(s) returned	0.000 sec / 0.000 sec
2	16:17:30	SELECT * FROM courses LIMIT 0, 1000	6 row(s) returned	0.000 sec / 0.000 sec
3	16:17:31	SELECT * FROM purchases LIMIT 0, 1000	12 row(s) returned	0.000 sec / 0.000 sec

Object Info Session

2. Data Exploration Using Joins

Data Presentation Guidelines for the following query

- Format currency values to 2 decimal places.
- Use aliases for column names (e.g., AS total_revenue).
- Sort results appropriately (e.g., highest total_spent first).

```
67 -- Format currency values to 2 decimal places. Use aliases for column names. Sort results appropriately.
68
69 • SELECT *, FORMAT(unit_price, 2) AS formatted_price
70 FROM courses;
71
72 • SELECT course_name, FORMAT(unit_price, 2) AS formatted_price
73 FROM courses;
```

course_id	course_name	category	Unit_Price	formatted_price
201	MySQL for Beginners	Database	1200.00	1,200.00
202	Advanced Excel	Analytics	900.00	900.00
203	Python for Data Analysis	Programming	1500.00	1,500.00
204	Power BI Dashboarding	Visualization	1800.00	1,800.00
205	Digital Marketing Basics	Marketing	1000.00	1,000.00
206	Statistics Essentials	Analytics	1300.00	1,300.00

Result 19 x

Output

Action Output

#	Time	Action	Message
1	16:25:19	use eleaming_platform	0 row(s) affected
2	16:26:07	SELECT *, FORMAT(unit_price, 2) AS formatted_price FROM courses LIMIT 0, 1000	6 row(s) returned

```
67 -- Format currency values to 2 decimal places. Use aliases for column names. Sort results appropriately.
68
69 • SELECT *, FORMAT(unit_price, 2) AS formatted_price
70 FROM courses;
71
72 • SELECT course_name, FORMAT(unit_price, 2) AS formatted_price
73 FROM courses;
```

course_name	formatted_price
MySQL for Beginners	1,200.00
Advanced Excel	900.00
Python for Data Analysis	1,500.00
Power BI Dashboarding	1,800.00
Digital Marketing Basics	1,000.00
Statistics Essentials	1,300.00

Result 20 x

Output

Action Output

#	Time	Action	Message
1	16:25:19	use eleaming_platform	0 row(s) affected
2	16:26:07	SELECT *, FORMAT(unit_price, 2) AS formatted_price FROM courses LIMIT 0, 1000	6 row(s) returned
3	16:28:23	SELECT course_name, FORMAT(unit_price, 2) AS formatted_price FROM courses LIMIT 0, 1000	6 row(s) returned

Use SQL INNER JOIN, LEFT JOIN, and RIGHT JOIN to:

- Combine learner, course, and purchase data.
- Display each learner's purchase details (course name, category, quantity, total amount, and purchase date).

```
75 -- Use SQL INNER JOIN, LEFT JOIN, and RIGHT JOIN to:
76 -- Combine learner, course, and purchase data.
77 -- Display each learner's purchase details (course name, category, quantity, total amount, and purchase date).
78
79 • SELECT
80     l.full_name,
81     c.course_name,
82     c.category,
83     p.quantity,
84     CONCAT('₹', FORMAT(p.quantity * c.unit_price, 2)) AS total_amount,
85     p.purchase_date
86 FROM purchases p
87 INNER JOIN learners l
88     ON p.learner_id = l.learner_id
89 INNER JOIN courses c
90     ON p.course_id = c.course_id;
91
```

Result Grid

	full_name	course_name	category	quantity	total_amount	purchase_date
▶	Aarav Sharma	MySQL for Beginners	Database	1	₹1,200.00	2025-01-05
	Karthik Raj	MySQL for Beginners	Database	2	₹2,400.00	2025-01-20
	John Smith	Advanced Excel	Analytics	2	₹1,800.00	2025-01-10
	Priya Nair	Advanced Excel	Analytics	2	₹1,800.00	2025-01-28
	Priya Nair	Python for Data Analysis	Programming	1	₹1,500.00	2025-01-07
	Olivia Wilson	Python for Data Analysis	Programming	1	₹1,500.00	2025-01-22
	Emily Davis	Power BI Dashboarding	Visualization	1	₹1,800.00	2025-01-12
	Aarav Sharma	Power BI Dashboarding	Visualization	1	₹1,800.00	2025-01-25
	Rahul Verma	Digital Marketing Basics	Marketing	3	₹3,000.00	2025-01-15
	John Smith	Digital Marketing Basics	Marketing	1	₹1,000.00	2025-02-01
	Sophia Brown	Statistics Essentials	Analytics	1	₹1,300.00	2025-01-18
	Rahul Verma	Statistics Essentials	Analytics	2	₹2,600.00	2025-02-04

Result 22 x

Output

Action Output

#	Time	Action	Message
1	16:31:05	SELECT l.full_name, c.course_name, c.category, p.quantity, CONCAT('₹', FORMAT(p.quantity * c.u...	12 row(s) returned

3. Analytical Queries:

Q1. Display each learner's total spending (quantity × unit_price) along with their country.

The screenshot shows a web application window titled "E-Learning Platform - 1". The interface includes a toolbar with icons for file operations, a "Limit to 1000 rows" dropdown, and a search icon. The main area displays a SQL query for Q1, which calculates the total spending for each learner by joining the learners, purchases, and courses tables. The query uses a subquery to calculate the total spending for each learner and then groups the results by learner_id, full_name, and country.

```
-- Q1. Display each learner's total spending (quantity × unit_price) along with their country.
SELECT
  l.learner_id,
  l.full_name,
  l.country,
  CONCAT('₹', FORMAT(SUM(p.quantity * c.unit_price), 2)) AS total_spending
FROM learners l
JOIN purchases p
  ON l.learner_id = p.learner_id
JOIN courses c
  ON p.course_id = c.course_id
GROUP BY l.learner_id, l.full_name, l.country;
```

Below the query editor, the "Result Grid" is displayed, showing a table with 4 columns: learner_id, full_name, country, and total_spending. The table contains 8 rows of data, with the first row highlighted. The total_spending values are formatted with a rupee symbol and two decimal places.

learner_id	full_name	country	total_spending
101	Aarav Sharma	India	₹3,000.00
107	Karthik Raj	India	₹2,400.00
103	John Smith	USA	₹2,800.00
102	Priya Nair	India	₹3,300.00
108	Olivia Wilson	Australia	₹1,500.00
104	Emily Davis	UK	₹1,800.00
105	Rahul Verma	India	₹5,600.00
106	Sophia Brown	Canada	₹1,300.00

At the bottom of the interface, the "Result 23" tab is active, showing the "Output" section. The "Action Output" dropdown is set to "SELECT". The output table shows the execution details, including the time (16:49:15), the action (SELECT), and the message (8 row(s) returned).

#	Time	Action	Message
1	16:49:15	SELECT	8 row(s) returned

Q2. Find the top 3 most purchased courses based on total quantity sold.

The screenshot displays the E-Learning Platform interface. At the top, a toolbar includes icons for file operations and a 'Limit to 1000 rows' dropdown. The main area shows a SQL query editor with the following code:

```
108 -- Q2. Find the top 3 most purchased courses based on total quantity sold.
109
110 • SELECT *
111 FROM (
112     SELECT
113         c.course_id,
114         c.course_name,
115         SUM(p.quantity) AS total_quantity_sold,
116         DENSE_RANK() OVER (ORDER BY SUM(p.quantity) DESC) AS course_rank
117     FROM courses c
118     JOIN purchases p
119     ON c.course_id = p.course_id
120     GROUP BY c.course_id, c.course_name
121 ) ranked_courses
122 WHERE course_rank <= 3;
```

Below the query editor is a 'Result Grid' section with a 'Filter Rows' input and an 'Export' button. The grid contains the following data:

course_id	course_name	total_quantity_sold	course_rank
202	Advanced Excel	4	1
205	Digital Marketing Basics	4	1
201	MySQL for Beginners	3	2
206	Statistics Essentials	3	2
203	Python for Data Analysis	2	3
204	Power BI Dashboarding	2	3

At the bottom, the 'Output' section shows 'Action Output' with a table of execution details:

#	Time	Action	Message
1	17:12:41	SELECT * FROM (SELECT c.course_id, c.course_name, SUM(p.quantity) AS total_quantity_s...	6 row(s) returned

Q3. Show each course category's total revenue and the number of unique learners who purchased from that category.

The screenshot shows a web application interface for a SQL query editor. The query is as follows:

```
124
125 -- Q3. Show each course category's total revenue and the number of unique learners who purchased from that category.
126
127 • SELECT
128     c.category,
129     CONCAT('₹', FORMAT(SUM(p.quantity * c.unit_price), 2)) AS total_revenue,
130     COUNT(DISTINCT p.learner_id) AS unique_learners
131 FROM courses c
132 JOIN purchases p
133     ON c.course_id = p.course_id
134 GROUP BY c.category;
```

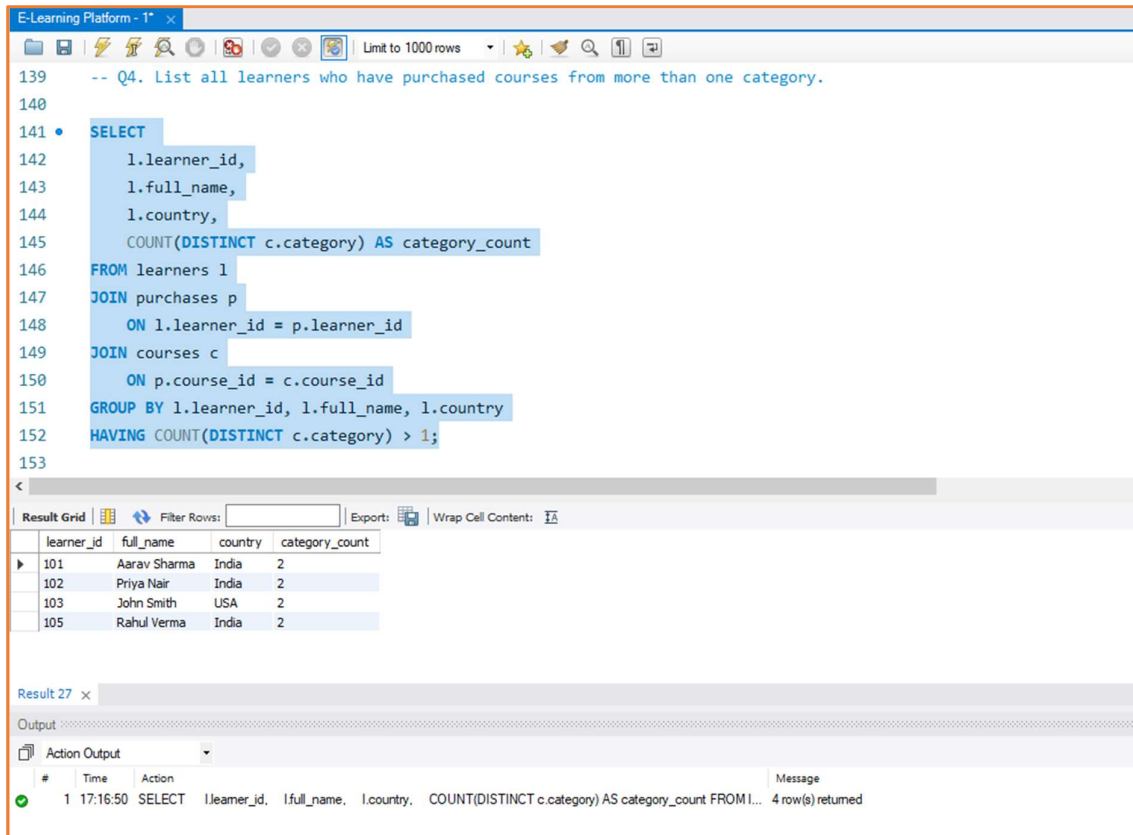
Below the query editor, the results are displayed in a table with the following data:

category	total_revenue	unique_learners
Analytics	₹7,500.00	4
Database	₹3,600.00	2
Marketing	₹4,000.00	2
Programming	₹3,000.00	2
Visualization	₹3,600.00	2

At the bottom, the 'Action Output' section shows the execution details:

#	Time	Action	Message
1	17:15:08	SELECT c.category, CONCAT('₹', FORMAT(SUM(p.quantity * c.unit_price), 2)) AS total_revenue, COUNT...	5 row(s) returned

Q4. List all learners who have purchased courses from more than one category.



```
139 -- Q4. List all learners who have purchased courses from more than one category.
140
141 • SELECT
142     l.learner_id,
143     l.full_name,
144     l.country,
145     COUNT(DISTINCT c.category) AS category_count
146 FROM learners l
147 JOIN purchases p
148     ON l.learner_id = p.learner_id
149 JOIN courses c
150     ON p.course_id = c.course_id
151 GROUP BY l.learner_id, l.full_name, l.country
152 HAVING COUNT(DISTINCT c.category) > 1;
153
```

Result Grid

learner_id	full_name	country	category_count
101	Aarav Sharma	India	2
102	Priya Nair	India	2
103	John Smith	USA	2
105	Rahul Verma	India	2

Result 27 x

Output

Action Output

#	Time	Action	Message
1	17:16:50	SELECT l.learner_id, l.full_name, l.country, COUNT(DISTINCT c.category) AS category_count FROM l...	4 row(s) returned

Q5. Identify courses that have not been purchased at all.

The screenshot shows a web application window titled "E-Learning Platform - 1* x". The interface includes a toolbar with icons for file operations, a "Limit to 1000 rows" dropdown, and a search icon. The main area is a SQL editor with line numbers 154 to 168. The SQL query is as follows:

```
154
155
156 -- Q5. Identify courses that have not been purchased at all.
157
158 • SELECT
159     c.course_id,
160     c.course_name,
161     c.category,
162     c.unit_price
163 FROM courses c
164 LEFT JOIN purchases p
165     ON c.course_id = p.course_id
166 WHERE p.purchase_id IS NULL;
167
168
```

Below the editor is a "Result Grid" section with a "Filter Rows:" input and "Export" and "Wrap Cell Content" options. The grid has columns for "course_id", "course_name", "category", and "unit_price". Below this is an "Output" section with a tab labeled "Action Output". The output table has columns for "#", "Time", "Action", and "Message". The first row shows a successful execution of the query at 17:18:39, returning 0 rows.

#	Time	Action	Message
1	17:18:39	SELECT c.course_id, c.course_name, c.category, c.unit_price FROM courses c LEFT JOIN purchase...	0 row(s) returned